

NEREID-B

Nonlinear Eulerian Reactive-osmotic Effluent Integro-Dispersion model for Brine outfalls

A Novel Unsteady, Stochastic, Coupled Hydro–Haline–Thermal Partial-Differential-
Equation System for Predicting the Three-Dimensional, Time-Evolving Salinity
Distribution of a Negatively-Buoyant Brine Plume Discharged from a Submarine Pipe
into a Moving Sea

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Abstract

NEREID-B is a fully-coupled, unsteady, non-Boussinesq, stochastic partial-differential-equation (PDE) system that predicts the three-dimensional space-time evolution of salinity produced when a concentrated brine effluent (e.g. the reject stream of a seawater reverse-osmosis desalination plant) is discharged through an inclined submarine pipe into a moving, stratified, wave- and tide-forced sea. The model simultaneously solves seven tightly-coupled fields — the velocity vector, pressure, density, salinity, temperature, turbulence state, and the air-sea interface position — and closes them with a nonlinear equation of state, an irreversible-thermodynamics (Onsager) cross-diffusion law that couples heat and salt, and an explicit osmotic-pressure body force derived from the reverse-osmosis thermodynamics of the discharged reject.

Three features make the formulation novel and patentable: (i) the osmotic-pressure gradient of the hyper-saline reject is promoted to a first-class momentum and mass-flux term rather than being neglected as in all conventional plume models; (ii) the unpredictability of the sea, wind and turbulence is represented intrinsically as space-time colored-noise stochastic forcing, so that the model returns a full probability density of the salinity field — a predicted mean plume together with a quantified confidence envelope of how far and how deep the brine reaches — rather than a single deterministic answer; and (iii) salinity and temperature are coupled through a complete anisotropic, state-dependent dispersion tensor that folds together molecular diffusion, shear dispersion, wave stirring, and bathymetric steering. The model is posed on a terrain-following coordinate that resolves arbitrary bathymetry and continental-shelf geometry, and is closed by a discharge boundary condition that embeds the full pipe hydraulics, nozzle inclination, and splash-zone air-sea exchange.

1. Problem Statement, Scope and Novelty

1.1 Physical problem

A submarine outfall pipe of length L_p , internal diameter d_p and terminal inclination θ discharges a brine effluent of salinity S_0 (typically $55\text{--}80\text{ g kg}^{-1}$ for SWRO reject, against an ambient of $\approx 35\text{--}39\text{ g kg}^{-1}$) and momentum into a receiving sea. Because the reject is denser than the ambient, the discharge forms a negatively-buoyant jet (a 'dense jet' or 'inverted plume') that rises, bends over under gravity, falls back, impacts the seabed, and then spreads as a stratified gravity current that hugs the bathymetry. The sea is never static: currents, tides, waves, wind-driven shear, vortices and turbulence continuously advect and dilute the plume, and the air-sea interface (the splash zone) exchanges momentum, heat and freshwater (evaporation) with the water column. The engineering questions are: at every instant, what is the salinity at every point; how far horizontally and how deep does the elevated-salinity field extend; and — because the sea is unpredictable — with what probability is a regulatory salinity threshold exceeded at a given location?

1.2 Why existing models are insufficient

- Integral/entrainment jet models (e.g. CORMIX, VISJET, JETLAG) collapse the plume to a 1-D centerline and cannot resolve the 3-D unsteady field, bathymetric steering, or the full sea state.
- Boussinesq Reynolds-Averaged Navier-Stokes (RANS) models assume small density differences and therefore lose accuracy for hyper-saline reject, where $(\rho - \rho_0)/\rho_0$ can exceed 0.04.
- All conventional models neglect osmotic pressure and the irreversible-thermodynamics coupling of heat and salt, and treat the sea deterministically, giving no measure of predictive uncertainty.

1.3 Statement of novelty (patentable claims)

NEREID-B advances the state of the art through the following coupled innovations, each of which is, to the authors' knowledge, absent from prior salinity-dispersion models:

1. Osmotic-hydrodynamic coupling: the gradient of the osmotic pressure $\Pi(S,T)$ of the hyper-saline field enters the momentum equation as a thermodynamic body force and the salt equation as an osmotic flux, capturing reverse-osmosis-driven micro-transport at the brine front.
2. Intrinsic stochastic forcing: ocean currents, tides, wind and sub-grid turbulence are represented by Ornstein–Uhlenbeck colored-noise PDE forcing, so the model is a genuine stochastic PDE (SPDE) whose solution is the probability density functional of the salinity field, $P[S(x,t)]$.
3. Non-Boussinesq variable-density formulation valid for arbitrarily large brine/ambient density contrast.
4. Onsager cross-diffusion: Soret (thermo-diffusion) and Dufour (diffusion-thermo) terms couple the salt and heat equations beyond their density coupling.
5. Fully anisotropic, state-dependent dispersion tensor D_S that fuses molecular diffusion, turbulent diffusion, shear (Taylor) dispersion, wave-orbital stirring and bathymetric anisotropy.
6. Embedded pipe-hydraulics and splash-zone boundary operators that couple the internal pipe flow, the inclined nozzle jet, and the two-phase air-sea interface directly to the field equations.

2. Computational Domain, Coordinates and Bathymetry

Let $x = (x, y, z)$ be a right-handed Cartesian frame with z positive upward and $z = 0$ at the still-water level. The seabed is $z = -H(x, y)$, where H is the bathymetry; the continental-shelf and local terrain geometry are supplied as the field H and its slope ∇H . The instantaneous free surface is $z = \eta(x, y, t)$. To resolve sloping bathymetry and the moving surface exactly, the vertical coordinate is mapped to a terrain-following sigma coordinate $\sigma \in [-1, 0]$:

$$\sigma = (z - \eta) / (H + \eta), \text{ so that } \sigma = 0 \text{ at the surface and } \sigma = -1 \text{ at the bed.} \quad (1)$$

All field equations below are written in physical (x, y, z) form for clarity; the model is integrated in the (x, y, σ, t) image where the domain is a fixed prism. The total water depth is $D(x, y, t) = H + \eta$. The pipe terminus (the diffuser nozzle) is located at $x_d = (x_d, y_d, z_d)$ with $z_d = -H(x_d, y_d) + h_n$, h_n being the nozzle height above the bed.

3. The Coupled Governing PDE System

The state vector solved by NEREID-B is

$$q(x, t) = (\rho, u = (u, v, w), p, S, T, k, \varepsilon, \eta, \alpha)^T, \quad (2)$$

comprising density ρ , velocity u , pressure p , absolute salinity S , conservative temperature T , turbulent kinetic energy k , its dissipation ε , the free-surface elevation η , and the air-water volume fraction α . The governing balance laws follow.

3.1 Conservation of mass (non-Boussinesq continuity)

Because the brine/ambient density contrast is not small, mass is conserved in fully compressible-in-density (non-Boussinesq) form:

$$\partial \rho / \partial t + \nabla \cdot (\rho u) = 0. \quad (3)$$

The incompressible Boussinesq limit $\nabla \cdot u = 0$ is recovered automatically where $|\rho - \rho_0| / \rho_0 \ll 1$, but is never assumed.

3.2 Momentum balance (Reynolds-averaged Navier–Stokes with buoyancy, rotation, waves, osmosis and stochastic forcing)

$$\partial(\rho u) / \partial t + \nabla \cdot (\rho u \otimes u) = -\nabla p + \nabla \cdot \tau + \rho g - 2\rho \Omega \times u + F_{\text{wave}} + F_{\text{osm}} + F_{\text{stoch}}. \quad (4)$$

Term by term:

- Deviatoric + turbulent (Reynolds) stress, with eddy viscosity μ_t :

$$\tau = (\mu + \mu_t)(\nabla u + \nabla u^T - (2/3)(\nabla \cdot u) I) - (2/3) \rho k I. \quad (5)$$

- Buoyancy: ρg with $g = -g \hat{z}$; the active driver of the dense plume because ρ depends on S and T through the equation of state (§4.1).
- Coriolis (Earth rotation): $-2\rho \Omega \times u$, $\Omega = \Omega(0, \cos \phi, \sin \phi)$, ϕ = latitude.
- Wave forcing as the divergence of the radiation-stress tensor S_{rad} (depth-dependent Longuet-Higgins / vortex-force form):

$$F_{\text{wave}} = -\nabla \cdot S_{\text{rad}} + \rho (u_{\text{St}} \times \omega), \quad u_{\text{St}} = \text{Stokes drift}, \quad \omega = \nabla \times u. \quad (6)$$

- Osmotic body force (NOVEL) — the gradient of osmotic pressure Π of the saline field acts on the fluid through the partial molar volume of water $\bar{v}_w$:

$$F_{\text{osm}} = -(\varphi_v / \bar{v}_w) \nabla \Pi(S, T), \quad \varphi_v = \text{local solute volume fraction} . \quad (7)$$

- Stochastic forcing F_{stoch} (NOVEL) — see §3.7; represents the unresolved/unpredictable momentum input from gusts, eddies and internal waves.

3.3 Absolute-salinity transport (advection – anisotropic dispersion – osmotic & Soret flux – reaction – noise)

Salinity is transported by advection and by the divergence of a generalized flux J_S built from Onsager irreversible thermodynamics:

$$\partial S / \partial t + u \cdot \nabla S = -(1/\rho) \nabla \cdot J_S + R_S + \xi_S, \quad (8)$$

$$J_S = -\rho D_{S \cdot} \nabla S - \rho D_{ST} \nabla T - \rho (D_S L_p / R_g T) \nabla \Pi . \quad (9)$$

The three flux contributions are, respectively: anisotropic dispersion ($D_{S \cdot}$, a full tensor, §4.2); Soret thermo-diffusion D_{ST} that lets temperature gradients drive salt (cross-coupling to §3.4); and the osmotic/reverse-osmosis flux (NOVEL), in which the osmotic-pressure gradient drives an additional salt flux with permeability coefficient L_p — the open-water analogue of reverse-osmosis transport across the brine front. R_S is any reactive/source term (normally 0 for salt; retained for generality, e.g. anti-scalant tracers), and ξ_S is scalar stochastic forcing.

3.4 Conservative-temperature (heat) transport with Dufour coupling

$$\partial T / \partial t + u \cdot \nabla T = -(1/\rho c_p) \nabla \cdot J_Q + Q_{\text{rad}}/(\rho c_p) + \xi_T, \quad (10)$$

$$J_Q = -\rho c_p D_{T \cdot} \nabla T - \rho c_p D_{TS} \nabla S . \quad (11)$$

Here $D_{T \cdot}$ is the anisotropic thermal dispersion tensor, D_{TS} is the Dufour (diffusion-thermo) coefficient that closes the reciprocal Onsager pair with D_{ST} , Q_{rad} is solar/longwave radiative heating, and ξ_T is thermal stochastic forcing. Equations (3.3) and (3.4) are thus two-way coupled both through the density (buoyancy) and directly through the cross-diffusion pair (D_{ST} , D_{TS}).

3.5 Nonlinear equation of state (haline–thermal–baric closure)

Density is closed by the full nonlinear TEOS-10 / Gibbs-function equation of state — not a linearized approximation — because the dense plume's dynamics are exquisitely sensitive to small density errors:

$$\rho = \rho_{\text{EOS}}(S, T, p) = \rho_0 [1 + \beta_S(S-S_0) - \alpha_T(T-T_0) + \gamma_p(p-p_0) + \Phi_{\text{nl}}] , \quad (12)$$

where α_T is the thermal-expansion, β_S the haline-contraction and γ_p the compressibility coefficient, all themselves functions of (S, T, p) , and Φ_{nl} collects the cabbeling/thermobaric nonlinear terms of TEOS-10. This single relation is the master coupling that ties the momentum, salinity and temperature equations together.

3.6 Turbulence closure (buoyancy-modified, stratification-damped k-ε)

Turbulent mixing — which controls dilution — is closed by a two-equation k-ε model modified for buoyancy so that the stable stratification of the dense plume correctly suppresses vertical mixing:

$$\partial(\rho k)/\partial t + \nabla \cdot (\rho u k) = \nabla \cdot [(\mu + \mu_t/\sigma_k)\nabla k] + P_k + G_b - \rho \varepsilon, \quad (13)$$

$$\partial(\rho \varepsilon)/\partial t + \nabla \cdot (\rho u \varepsilon) = \nabla \cdot [(\mu + \mu_t/\sigma_\varepsilon)\nabla \varepsilon] + (\varepsilon/k)[C_1(P_k + C_3 G_b) - C_2 \rho \varepsilon], \quad (14)$$

$$\mu_t = \rho C_\mu k^2/\varepsilon, \quad P_k = \tau : \nabla u, \quad G_b = (\mu_t/\rho Sc_t) g \cdot \nabla \rho. \quad (15)$$

The buoyancy-production term G_b is negative in stable stratification, damping turbulence exactly where the brine layer is densest — the physical mechanism that makes brine pool on the seabed. Standard constants ($C_\mu=0.09$, $C_1=1.44$, $C_2=1.92$, $\sigma_k=1.0$, $\sigma_\varepsilon=1.3$, $Sc_t=0.7$) are defaults; C_3 controls stratified mixing.

3.7 Stochastic forcing of an unpredictable sea (the SPDE layer)

The sea, wind and turbulence are not deterministic. NEREID-B promotes the system to a stochastic PDE by driving the momentum and scalar equations with space-time colored noise generated by Ornstein–Uhlenbeck (OU) processes, one per forcing channel m (currents, tide phase, wind, sub-grid turbulence):

$$\partial \zeta_m / \partial t = -\zeta_m / \tau_m + \sigma_m \mathcal{L}_m^{1/2} \dot{W}_m(x,t), \quad (16)$$

where τ_m is the de-correlation time, σ_m the intensity, $\mathcal{L}_m$ a spatial smoothing (covariance) operator giving the noise a realistic spatial correlation length ℓ_m , and $\dot{W}_m$ space-time white noise. The forcings in §3.2–3.4 are then

$$F_{\text{stoch}} = \rho \Sigma_m a_m \zeta_m, \quad \xi_S = \Sigma_m b_m \zeta_m \cdot \nabla S, \quad \xi_T = \Sigma_m c_m \zeta_m \cdot \nabla T. \quad (17)$$

Solving the ensemble (Monte-Carlo over realizations of $\dot{W}_m$, or the associated Fokker–Planck moment equations) yields the predictive probability density of the salinity field:

$$P[S(x,t)], \text{ with mean } \langle S \rangle, \text{ variance } \sigma_S^2, \text{ and exceedance probability } \mathbb{P}(S(x,t) > S_{\text{crit}}). \quad (18)$$

This is the formal mechanism by which NEREID-B 'captures unpredictability': it predicts not a single number but a calibrated distribution of how far and how deep the brine plume may reach.

4. Constitutive and Closure Relations

4.1 Osmotic pressure (Pitzer-corrected van 't Hoff)

The osmotic pressure that drives F_{osm} and the osmotic salt flux is computed from a van 't Hoff law with an activity (osmotic) coefficient $\phi_{\text{os}}(S,T)$ that corrects for the strong non-ideality of concentrated brine (Pitzer model):

$$\Pi(S,T) = \varphi_{os}(S,T) \cdot v \cdot (\rho_w S / M_s) \cdot R_g T, \quad (19)$$

with v the number of dissolved ions per formula unit of salt, M_s the mean molar mass of the dissolved salts, R_g the gas constant and ρ_w the water density. For seawater $\Pi \approx 2.7$ MPa at $S = 35$ g kg⁻¹ and rises steeply for reject brine, so $\nabla \Pi$ across the brine front is dynamically non-negligible — the basis of the osmotic coupling.

4.2 The anisotropic, state-dependent dispersion tensor (NOVEL closure)

The salt dispersion tensor fuses every relevant mixing mechanism into a single symmetric positive-definite tensor aligned with the local flow, shear, wave and bathymetric directions:

$$D_S = D_{mol} I + D_{turb}(k,\varepsilon) I + D_{shear}(|\nabla u|) \hat{e}_u \otimes \hat{e}_u + D_{wave}(H_s, T_w) \hat{e}_w \otimes \hat{e}_w + D_{bath}(\nabla H) (I - \hat{n} \otimes \hat{n}). \quad (20)$$

The five contributions are molecular diffusion (D_{mol}), isotropic turbulent diffusion ($D_{turb} = \mu_t / \rho Sc_t$), longitudinal shear/Taylor dispersion along the flow unit-vector $\hat{e}_u$, wave-orbital stirring along the wave direction $\hat{e}_w$ (a function of significant wave height H_s and period T_w), and an along-slope enhancement tangent to the bed ($\hat{n}$ = bed normal) that steers the dense gravity current along the bathymetry. The thermal tensor D_T has the identical structure with the thermal molecular diffusivity and turbulent Prandtl number.

5. Boundary and Initial Conditions

5.1 Discharge (diffuser-nozzle) condition with embedded pipe hydraulics

At the nozzle Γ_d the jet velocity, direction and salinity are imposed. The exit speed U_d and pressure are obtained from the internal pipe hydraulics (Darcy–Weisbach) so that pipe length, diameter, roughness and inclination feed the field directly:

$$U_d = Q_d / (\pi d_p^2 / 4), \quad p_d = p_{man} - \rho_b g \Delta z_p - (f L_p / d_p + \sum K_{minor}) (\rho_b U_d^2 / 2), \quad (21)$$

$$u|_{\Gamma_d} = U_d (\cos\theta \cos\psi, \cos\theta \sin\psi, \sin\theta), \quad S|_{\Gamma_d} = S_0, \quad T|_{\Gamma_d} = T_b, \quad (22)$$

where Q_d is the discharge flow rate, f the Darcy friction factor (roughness-/Reynolds-dependent), Δz_p the pipe rise, K_{minor} the minor-loss coefficients, p_{man} the manifold pressure, ρ_b the brine density, θ the nozzle elevation (inclination) angle and ψ its azimuth. The jet momentum flux $M_o = \rho_b U_d^2 A_d$ and buoyancy flux $B_o = g(\rho_b - \rho_a) / \rho_a \cdot Q_d$ set the densimetric Froude number that governs the trajectory.

5.2 Free surface and splash zone (two-phase air–sea coupling)

The air–water interface is tracked by a volume-of-fluid fraction α ($\alpha=1$ water, 0 air), advected with the flow, which resolves the splash zone and surface deformation:

$$\partial \alpha / \partial t + \nabla \cdot (\alpha u) = 0, \quad \rho = \alpha \rho_{water} + (1 - \alpha) \rho_{air}. \quad (23)$$

Across the interface the dynamic and kinematic conditions, wind stress, and the evaporative salinity flux apply:

$$\partial\eta/\partial t + u_h \cdot \nabla\eta = w, \quad (\text{kinematic}) \quad (24)$$

$$\tau_{\text{surf}} = \rho_a C_d |U_{10} - u_h|(U_{10} - u_h), \quad (\text{wind stress, } U_{10} = 10\text{-m wind}) \quad (25)$$

$$(\rho D_S \nabla S) \cdot \hat{n} = S(E - P)/\rho_w, \quad (\text{evaporation } E \text{ minus precipitation } P \text{ concentrates salt}). \quad (26)$$

The air velocity / wind field $U_{10}(x,t)$ is itself one of the stochastic OU channels of §3.7, so wind unpredictability propagates into the surface stress and the splash-zone mixing.

5.3 Seabed and shoreline

$$u \cdot \hat{n} = 0 \quad (\text{no-normal-flow}), \quad \text{wall-function } \tau_{\text{bed}} = \rho C_D^{\text{bed}} |u_b| u_b, \quad (D_S \nabla S) \cdot \hat{n} = 0 \quad (\text{no salt flux through bed}). \quad (27)$$

The bed-normal $\hat{n}$ follows the bathymetry/terrain ∇H , so the dense gravity current is steered along the real continental-shelf slope.

5.4 Open-ocean lateral boundaries (currents, tides)

Far-field boundaries impose the ambient stratification and the current/tidal forcing, again with a stochastic component:

$$u|_{\infty} = U_{\text{current}}(z,t) + U_{\text{tide}}(t) + \zeta_{\text{current}}, \quad S|_{\infty} = S_{\text{amb}}(z), \quad T|_{\infty} = T_{\text{amb}}(z), \quad (28)$$

$$U_{\text{tide}}(t) = \sum_j A_j \cos(\omega_j t + \varphi_j) \quad (\text{harmonic tidal constituents}). \quad (29)$$

5.5 Initial condition

$$q(x,0) = q_{\text{amb}}(x): \text{ quiescent or measured ambient profiles of } S, T, \rho, u; \text{ plume fields seeded at the nozzle.} \quad (30)$$

6. Coupling Structure

The seven balance laws are not solved in isolation; they form a single coupled operator. The principal coupling pathways are:

- $\rho \leftrightarrow (S,T,p)$: the nonlinear EOS (3.5) — salt and heat set density, density sets buoyancy in the momentum equation (3.2).
- $u \leftrightarrow (S,T)$: velocity advects salt and heat (3.3, 3.4); their gradients set buoyancy that drives velocity (3.2).
- $S \leftrightarrow T$: direct Onsager cross-diffusion (Soret D_{ST} , Dufour D_{TS}) beyond the density route.
- $(k,\epsilon) \leftrightarrow$ all: turbulence sets μ_t , D_{turb} in every equation; mean shear and buoyancy gradients set k and ϵ (3.6).

- $\Pi \leftrightarrow (u, S)$: osmotic pressure of the saline field forces momentum and salt flux (F_{osm}, J_S) — the novel osmotic loop.
- $\zeta_m \leftrightarrow \text{all}$: stochastic channels perturb momentum, scalars and boundaries, turning the deterministic PDE into an SPDE.

7. Predicted Outputs (Diagnostic Functionals)

From the solved fields, NEREID-B reports the engineering quantities of interest, each as a time series and as a probability distribution:

$$\text{Dilution: } \mathcal{D}(x, t) = (S_0 - S_{\text{amb}}) / (S(x, t) - S_{\text{amb}}), \quad (31)$$

$$\text{Excess-salinity footprint: } \mathcal{A}(t) = \iint \mathbb{1}[S(x, y, z^*, t) - S_{\text{amb}} > \Delta S_{\text{crit}}] dx dy, \quad (32)$$

$$\text{Maximum reach: } r_{\text{max}}(t) = \max\{|x_h - x_d| : S - S_{\text{amb}} > \Delta S_{\text{crit}}\}, \quad (33)$$

$$\text{Maximum depth of impact: } z_{\text{max}}(t) = \min\{z : S - S_{\text{amb}} > \Delta S_{\text{crit}}\}, \quad (34)$$

$$\text{Seabed impact (return) point and the gravity-current run-out distance,} \quad (35)$$

$$\text{Exceedance risk: } \mathbb{R}(x, t) = \mathbb{P}[S(x, t) - S_{\text{amb}} > \Delta S_{\text{crit}}] \text{ from the SPDE ensemble.} \quad (36)$$

ΔS_{crit} is the regulatory salinity-increment limit (commonly +2 to +5 % of ambient, or a fixed +2 g kg⁻¹ above background). The pair $(r_{\text{max}}, z_{\text{max}})$ with its confidence envelope is the direct answer to 'how far and how deep does the brine go'.

8. Governing Dimensionless Groups

Non-dimensionalizing the system reveals the controlling parameters, useful for scaling, validation and reduced-order use:

Group	Definition	Physical role
Fr_d	$U_d / \sqrt{g' d_p}$	Densimetric Froude — jet momentum vs buoyancy; sets rise height
g'	$g(\rho_b - \rho_a) / \rho_a$	Reduced gravity of the reject
Re	$\rho U_d d_p / \mu$	Reynolds — turbulence onset
Ri	$g' / (\partial u / \partial z)^2 \cdot \partial \rho / \partial z$	Richardson — stratified mixing suppression
Pe	$U_d d_p / D_{\text{mol}}$	Péclet — advection vs diffusion of salt
Sc	ν / D_{mol}	Schmidt — momentum vs salt diffusivity
Ra	$g' L^3 / (\nu D_S)$	Rayleigh — buoyant convection strength
Ro	$U / (f L)$	Rossby — rotation (Coriolis) importance
Π^*	$\Pi / (\rho U_d^2)$	Osmotic-to-inertial pressure ratio (novel group)
K_u	$\sigma_m \tau_m / U$	Stochastic-forcing intensity (novel group)

9. Complete Parameter Inventory (Model Inputs)

Every quantity the model ingests, grouped by source. Symbols match the equations above. These are the inputs that must be supplied or measured to drive NEREID-B.

9.1 Effluent / brine discharge

Symbol	Parameter	Units
S_0	Brine (effluent) absolute salinity	g kg^{-1}
Q_d	Volumetric discharge rate	$\text{m}^3 \text{s}^{-1}$
U_d	Nozzle exit velocity	m s^{-1}
p_d	Discharge pressure at nozzle	Pa
T_b	Brine temperature	$^{\circ}\text{C}$
ρ_b	Brine density	kg m^{-3}
θ, ψ	Nozzle inclination & azimuth angles	deg
S_{anti}	Anti-scalant / additive tracer conc.	mg L^{-1}

9.2 Pipe / diffuser geometry & hydraulics

Symbol	Parameter	Units
L_p	Pipe length	m
d_p	Pipe / port internal diameter	m
ϵ_r	Pipe wall roughness	m
f	Darcy friction factor	—
N_{port}	Number of diffuser ports	—
s_{port}	Port spacing	m
h_n	Nozzle height above seabed	m
Δz_p	Pipe vertical rise	m
K_{minor}	Minor-loss coefficients	—

9.3 Ambient sea — physical state

Symbol	Parameter	Units
$S_{\text{amb}}(z)$	Ambient salinity profile	g kg^{-1}
$T_{\text{amb}}(z)$	Ambient temperature profile	$^{\circ}\text{C}$
$\rho_{\text{amb}}(z)$	Ambient density (stratification)	kg m^{-3}
$p(z)$	Hydrostatic + dynamic pressure	Pa
N^2	Brunt–Väisälä (stratification) frequency	s^{-2}
c_p	Specific heat capacity of seawater	$\text{J kg}^{-1} \text{K}^{-1}$
μ	Dynamic viscosity	Pa s
D_{mol}	Molecular salt diffusivity	$\text{m}^2 \text{s}^{-1}$

9.4 Ambient sea — dynamics (the moving sea)

Symbol	Parameter	Units
$U_{\text{current}}(z,t)$	Ocean current profile (magnitude & direction)	m s^{-1}
A_j, ω_j, φ_j	Tidal constituents (amp., freq., phase)	$\text{m, s}^{-1}, \text{rad}$
$H_s, T_w, \hat{e}_w$	Significant wave height, period, direction	m, s, —
u_{St}	Stokes drift (wave-induced)	m s^{-1}
$\omega = \nabla \times u$	Vorticity / vortex field	s^{-1}
φ_{lat}	Latitude (Coriolis parameter f)	deg
τ_m, σ_m, ℓ_m	Stochastic de-correlation time, intensity, length	s, —, m

9.5 Atmosphere / air–sea interface

Symbol	Parameter	Units
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U_{10}	Wind velocity at 10 m (air velocity)	m s^{-1}
C_d	Air–sea drag coefficient	–
ρ_a	Air density	kg m^{-3}
E, P	Evaporation, precipitation rates	m s^{-1}
Q_{rad}	Net radiative heat flux	W m^{-2}
T_{air}	Air temperature	$^{\circ}\text{C}$

9.6 Bathymetry, terrain & domain

Symbol	Parameter	Units
$H(x,y)$	Bathymetry / seabed depth	m
∇H	Bed slope (continental-shelf geometry)	–
C_D^{bed}	Bed friction coefficient	–
x_d	Discharge location	m
L_x, L_y, D	Domain extent & total depth	m

9.7 Thermodynamic / coupling coefficients

Symbol	Parameter	Units
$\alpha_T, \beta_S, \gamma_p$	Thermal-expansion, haline-contraction, compressibility	$\text{K}^{-1}, \text{kg g}^{-1}, \text{Pa}^{-1}$
ϕ_{os}	Osmotic (activity) coefficient	–
L_p	Osmotic permeability coefficient	s
D_{ST}, D_{TS}	Soret & Dufour cross-diffusion coefficients	$\text{m}^2 \text{s}^{-1} \text{K}^{-1}$ (eq.)
Sc_t, Pr_t	Turbulent Schmidt & Prandtl numbers	–
$\bar{v}_w, M_s, v$	Partial molar volume, salt molar mass, ion number	$\text{m}^3 \text{mol}^{-1}, \text{kg mol}^{-1}, \text{–}$

10. Recommended Numerical Solution Strategy

- Spatial discretization on a terrain-following σ -grid by a finite-volume method, refined near the diffuser and the bed; unstructured horizontal mesh to capture the coastline and bathymetry.
- Pressure–velocity coupling for variable density by a PISO/SIMPLE projection with the non-Boussinesq continuity constraint.
- Scalar transport (S, T) with a high-resolution, monotone (TVD) advection scheme to avoid spurious salinity over/undershoots.
- Semi-implicit time integration; CFL- and buoyancy-limited adaptive time step for the unsteady evolution from discharge to far field.
- Stochastic layer by Monte-Carlo ensemble (or polynomial-chaos / stochastic-Galerkin surrogate) over OU realizations to build $P[S(x,t)]$, the mean, variance and exceedance maps.
- Two-phase free surface by VOF (α) with the wind-stress and evaporation surface fluxes.
- Coupling solved by an outer Picard/Newton iteration each step until $\rho, u, S, T, k, \varepsilon$ are mutually converged.

11. Validation Plan and Well-Posedness

- Analytical limits: recover the classical $30^{\circ}/45^{\circ}/60^{\circ}$ inclined dense-jet trajectory and dilution scaling (Roberts et al.) when stochastic, osmotic and rotation terms are switched off.

- Laboratory benchmarks: compare rise height, return-point dilution and spreading-layer thickness against published dense-jet experiments.
- Field validation: CTD / ADCP transects of an operating SWRO outfall; verify the predicted $P[S]$ envelope contains the observed salinities at the stated confidence.
- Conservation checks: global salt and mass budgets closed to machine precision; entropy-production positivity of the Onsager fluxes guarantees thermodynamic consistency and dissipativity (a sufficient condition for well-posedness of the coupled parabolic-hyperbolic system).

12. Assumptions and Limitations

- Seawater is a Newtonian continuum; the reject carries no significant suspended-solid rheology (extendable via a particle-laden closure).
- The OU stochastic forcing is Gaussian and stationary over the forecast window; non-stationary storm forcing requires re-estimated $(\tau_m, \sigma_m, \ell_m)$.
- Chemical speciation is lumped into a single absolute-salinity scalar; multi-ion reverse-osmosis selectivity is a possible extension via Maxwell–Stefan multicomponent fluxes.
- Soret/Dufour and osmotic coefficients must be supplied for the specific brine chemistry; defaults are provided for NaCl-dominated SWRO reject.

13. Nomenclature (Principal Symbols)

Symbol	Meaning	Units
ρ, ρ_0	Density, reference density	kg m^{-3}
$\mathbf{u}=(\mathbf{u},\mathbf{v},\mathbf{w})$	Velocity vector	m s^{-1}
p	Pressure	Pa
S	Absolute salinity	g kg^{-1}
T	Conservative temperature	$^{\circ}\text{C} / \text{K}$
k, ϵ	Turbulent kinetic energy, dissipation	$\text{m}^2 \text{s}^{-2}, \text{m}^2 \text{s}^{-3}$
μ, μ_t	Molecular, turbulent (eddy) viscosity	Pa s
D_S, D_T	Salt, heat dispersion tensors	$\text{m}^2 \text{s}^{-1}$
Π	Osmotic pressure	Pa
$\mathbf{g}, \mathbf{g}'$	Gravity, reduced gravity	m s^{-2}
Ω, f	Rotation vector, Coriolis parameter	s^{-1}
η, α	Free-surface elevation, water fraction	m, –
S_{rad}	Wave radiation-stress tensor	N m^{-1}
ζ_m, ξ_S, ξ_T	Stochastic forcing channels & scalar noise	various
$F_{\text{osm}}, F_{\text{wave}}$	Osmotic, wave body forces	N m^{-3}
$\mathcal{D}, \mathcal{A}, r_{\text{max}}, z_{\text{max}}$	Dilution, footprint, reach, depth of impact	–, m^2 , m, m

14. Summary

NEREID-B unifies, for the first time in a single self-consistent PDE system, the non-Boussinesq hydrodynamics of a dense brine jet, the nonlinear haline-thermal-baric equation of state, irreversible-thermodynamics (Soret/Dufour) cross-diffusion, an explicit osmotic / reverse-osmosis coupling, a state-dependent anisotropic dispersion tensor, terrain-following bathymetry, two-phase splash-zone air-sea exchange, and an intrinsic stochastic-forcing layer that converts the model into a predictor of the full probability density of salinity. The result is a model that predicts, at every instant and with quantified uncertainty, the three-dimensional salinity field around a submarine outfall and precisely how far and how deep the discharged brine penetrates a moving, unpredictable sea — accurately, sustainably, and in a form suitable for patent protection.